

The raw materials, components, sensors, chips, parts and services such as machining, testing and assembly for ISRO satellites are all partially provided by private industries. Some of the industry partners of ISRO include L&T, Godrej and Sure Safety. The collaboration with Godrej goes all the way back to 1985, when Godrej provided the precision machining required for the titanium components of the Vikas engine, which is used in both the PSLV and GSLV rockets.

ISRO wants to take the participation of private companies to the next level. One of the oldest collaborators of ISRO is Bharat Electronics Limited, or BEL. BEL already makes solar panels and microwaves for Indian satellites, and now ISRO wants to outsource the building of entire satellites. Alpha Design Technologies, established as part of the "Make in India" initiative of, is a Bengaluru based company leading a consortium of private companies, tasked with building two complete satellites for India. Alpha Design Technologies was the first private company to co-assemble a satellite with ISRO, the navigation satellite IRNSS-1H. Unfortunately, the payload fairing did not open, and the satellite could not be deployed, despite the PSLV rocket performing as expected. It was the first mission in 24 years to fail, but fortunately they were no secondary payloads piggybacking on the ride. Private industries were responsible for the assembly, integration and testing of the IRNSS-1I navigation satellite, which was successfully launched by the PSLV-C41 mission in April in 2018.

TEAMINDUS

None of the teams met the deadline for the Google Lunar XPrize. The competition has now been modified to help the private spaceflight companies raise the funds they need to realise their moon missions. TeamIndus is still very much working on the mission to send a rover to the moon. Currently, there are plans to launch the mission in the second half of 2019, and the final dates will be announced once the launch services provider is decided. The launch services provider here means the organisation that will provide the

rocket for the launch. The mission is small enough to rideshare or piggyback along with other payloads.

Initially, the Japanese Hakuto was supposed to share an ISRO PSLV rocket with the TeamIndus mission. Now, the Hakuto rover will also be going on the TeamIndus lander. The extended mission timeline means that TeamIndus has more time to pack in even more experiments on the rover. The rover will not only demonstrate the technical capabilities of private Indian spaceflight, but will also be doing real science. Sheelika Ravishankar, Marketing Head of TeamIndus elaborates on the scientific payloads on board, "Our first lunar mission is carrying multiple scientific and commercial payloads. CNES (French Space Agency) is providing stereo HD cameras for our rover and we are also carrying Hakuto's rover Sorato onboard our lander. In terms of scientific payloads, we have three science experiments onboard the spacecraft. INFN has their MoonLIGHT-2 instrument that will use lunar ranging techniques to perform General Relativity tests. The Indian Institute of Astrophysics has their LUCI ultra-violet telescope onboard, while the InSTEM institute has an experiment studying regeneration of planarian worms in micro-gravity. Additionally, two student-built experiments are onboard, EARS and ZOI. EARS is a radiation shield tech demonstrator and the ZOI experiment aims to study the effects of radiation on cyanobacteria."

But that is not all. At the moment, everyone in TeamIndus is working dedicatedly on the first mission to the Moon. But TeamIndus definitely has plans to continue to be a pioneering Indian spaceflight company beyond the first mission. They plan to use the expertise and experience gained in the first mission for multiple future missions to the moon. Ravishankar explains, "Over the course of seven years, we have successfully completed qualification tests of our Spacecraft and rover ECA. At this point we can put together our flight ready Spacecraft in less than nine months and we will be ready to launch once the new launch provider is finalized. We are already a private space company building lunar

landers for space exploration. We are addressing the key elements of affordability, reliability and repeatability to make Space more accessible. We continue to enhance our capabilities and build capacity that will enable us to get to the Moon multiple times over the next 5 years. In the process we send back amazing science data from beyond the Earth-orbit."

THE OTHER ONES

Several other companies are contributing to India's emergence in the private space sector. A handful of others including Applied Research & Development Laboratories (ARDL), Astrome, Dhruva Space, Earth2Orbit Analytix and Xovian are also working towards the same goal of pushing the technological boundaries of what our country is capable of. For instance, ARDL has branched itself into four divisions including one, Antariksha Lab, dedicated to space research. They've come up with an entire kit for building a model satellite capable of being launched at a certain altitude as well. The company's objective is to establish an R&D presence related to various aspects of space technology.

Astrome Technologies is currently working on building very high bandwidth microsatellites to beam down internet from space. Using their patented "MM-Wave transponders", the company is aiming to deliver high speed broadband internet by 2020. Dealing primarily in small satellite launch systems and devices, Dhruva Space has collaborated with AMSAT India for the HAMSAT-II satellite launch, and they have also collaborated with international organisations. Earth2Orbit Analytix takes care of advisory services to solve social, business and environmental problems and provide "location-aware, large-area analytics". Finally, Xovian takes care of providing "low cost sustainable solutions in satellite fabrication". Of course, there will be more companies which are already working towards building and researching space technology. Owing to ISRO's consistent success, far more companies will definitely see the light and aim to make bold new strides in the coming future. 